Biochemistry Cume A 2017-2018 Sakon

Physical-chemical characteristics of hen egg white proteins are shown below (Source: Powrie & Nakai, 1985). Use the table for the next 5 questions.

Protein	dry weight (%)	pI	MW (kDa)	Tm (°C)
Ovalbumin	54	4.5	44.5	84
Conalbumen	12	6.1	77.7	61
Ovomucoid	11	4.1	28	77
Ovomucin	3.5	4.5	6.0	60
Lysozyme	3.4	10.7	14.3	75
Globulin G2	4	5.5	49	92.5
Avidin	0.05	10	68.3	60

1. Homogenate of hen egg white was applied to a S-sepharose (strong cation exchanger) equilibrated using a buffer with pH=7. Some proteins did not bind to the column. The proteins were then eluted using increasing concentrations of NaCl at pH=7. What is the order of elution?

2. Homogenate of hen egg white was applied to a size exclusion chromatography equilibrated using a buffer with pH=7. The protein was then eluted using the same buffer. Which protein would you expect to elute last? What is the order of elution?

3. The relative electrophoretic mobility, under reducing conditions, of a 10 kDa protein and a 100 kDa protein are 0.90 and 0.10. Based on this, what is the relative mobility of avidin?

4. Which protein consists of 17 Lys+Arg residues and 9 Asp+Glu residues?

5. Propose the simplest method for the isolation of lysozyme from hen egg white homogenate.